Iterative Methods in Linear Algebra - Part 3
Introduction to Approximation Theory: Least Squares Method

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Convergence of Stationary Methods

stationary method:

$$\mathbf{x}^{(\mathbf{k}+1)} = \mathbf{x}^{(\mathbf{k})} + \mathbf{M}^{-1}\mathbf{r}^{(\mathbf{k})}$$

$\mathbf{x} \leftarrow$ exact solution of $\mathbf{Ax} = \mathbf{b}$
$\mathbf{r}^{(\mathbf{k})} = \mathbf{b} - \mathbf{Ax}^{(\mathbf{k})} \leftarrow$ (known) residual at iteration $\mathbf{k}$
$\mathbf{e}^{(\mathbf{k})} = \mathbf{x} - \mathbf{x}^{(\mathbf{k})} \leftarrow$ (unknown) error at iteration $\mathbf{k}$

goal: discuss the convergence of the method, that is, when does $\mathbf{e}^{(\mathbf{k})} \to \mathbf{0}$ when $\mathbf{k} \to \infty$?

Convergence of Stationary Methods

$$\mathbf{x} = (\mathbf{I} - \mathbf{M}^{-1}\mathbf{A})\mathbf{x} + \mathbf{M}^{-1}\mathbf{b};$$
$$\mathbf{x}^{(\mathbf{k}+1)} = (\mathbf{I} - \mathbf{M}^{-1}\mathbf{A})\mathbf{x}^{(\mathbf{k})} + \mathbf{M}^{-1}\mathbf{b};$$

subtract both equations to get

$$\mathbf{e}^{(\mathbf{k}+1)} = \mathbf{x} - \mathbf{x}^{(\mathbf{k}+1)} = (\mathbf{I} - \mathbf{M}^{-1}\mathbf{A})\mathbf{e}^{(\mathbf{k})} = \mathbf{T}\mathbf{e}^{(\mathbf{k})},$$

in which $\mathbf{T}$ is the iteration matrix.

Convergence of Stationary Methods

The solution of

$$\mathbf{e}^{(\mathbf{k}+\mathbf{1})} = \mathbf{T}\mathbf{e}^{(\mathbf{k})}$$

is

$$\mathbf{e}^{(\mathbf{k})} = \mathbf{T}^{\mathbf{k}}\mathbf{e}^{(\mathbf{0})},$$

in which $\mathbf{e}^{(\mathbf{0})} = \mathbf{x} - \mathbf{x}^{(\mathbf{0})}$ is a fixed initial error. Then, the method converges, that is, $\mathbf{e}^{(\mathbf{k})} \to \mathbf{0}$ when $\mathbf{k} \to \infty$ if and only if

$$\rho\left(\mathbf{T}\right) < 1.$$

Moreover, the smaller the $\rho\left(\mathbf{T}\right)$, the faster the convergence.

Convergence of Stationary Methods

- a sufficient condition for Jacobi and Gauss-Seidel to converge is that $\mathbf{A}$ is strictly diagonally dominant, that is,

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

  in this case, it can be shown that $\rho(\mathbf{I} - \mathbf{M}^{-1}\mathbf{A}) < 1$;
- if $\mathbf{A}$ is SPD, Gauss-Seidel converges; Jacobi may not converge on SPD matrices, but a slight modification to the iteration can guarantee convergence:

$$\rho(\mathbf{I} - \mathbf{w}\mathbf{D}^{-1}\mathbf{A}) < 1 \text{ for a suitable } 0 < w < 2/\lambda_{\max}(\mathbf{D}^{-1}\mathbf{A}).$$

Rate of Convergence

- how fast does the iterative method converge?
- how does $\mathbf{e}^{(\mathbf{k}+\mathbf{1})}$ compare to $\mathbf{e}^{(\mathbf{k})}$?

$$\mathbf{e}^{(\mathbf{k})} = \mathbf{T}^{\mathbf{k}}\mathbf{e}^{(\mathbf{0})} \Rightarrow \quad ||\mathbf{e}^{(\mathbf{k})}|| \approx ||\mathbf{T}^{\mathbf{k}}||\,||\mathbf{e}^{(\mathbf{0})}|| \Rightarrow \quad ||\mathbf{e}^{(\mathbf{k})}|| \approx \rho\left(\mathbf{T}\right)^{\mathbf{k}}||\mathbf{e}^{(\mathbf{0})}||;$$

of we take log on both sides:

$$\log||\mathbf{e}^{(\mathbf{k})}|| \approx k\log\rho\left(\mathbf{T}\right) + \log||\mathbf{e}^{(\mathbf{0})}||,$$

which means that the plot of the log of the error vs. k is a straight line with a negative slope.

Rate of Convergence

- how many iterations until the error norm drops an order of magnitude from $||\mathbf{e}^{(0)}||$?
- that is, how many iterations are needed to reduce the error norm by a factor of 10?

$$||\mathbf{e}^{(\mathbf{k})}|| \approx 0.1||\mathbf{e}^{(\mathbf{0})}|| \approx \rho\left(\mathbf{T}\right)^{\mathbf{k}} ||\mathbf{e}^{(\mathbf{0})}||;$$

take the $\log_{10}$ and solve for k:

$$k = -\frac{1}{\log \rho\left(\mathbf{T}\right)} = \frac{1}{\text{rate}} \quad \Rightarrow \quad \text{rate} = -\log_{10} \rho\left(\mathbf{T}\right).$$

Rate of Convergence and Spectral Radius

- the rate of convergence depends on $\rho(\mathbf{T})$, the spectral radius of the iteration matrix;
- there are reasonable methods we may use to estimate $\lambda_{\max}$ of $\mathbf{I} - \mathbf{M}^{-1}\mathbf{A}$;
- when we know where $\mathbf{A}$ is coming from, we have some *a-priori* estimates: for example, when $\mathbf{A}$ is a discretization of Poisson's equation on a regular grid of spacing h, then

$$\rho(\mathbf{T}) = 1 - ch^2,$$

and as discretization gets more refined, $h \to 0$, convergence becomes very slow.

Termination Criterion: Criterion 1

What are reasonable criteria to terminate the iteration?

A common criterion is to define `tol` and stop when

$$||\mathbf{r}^{(\mathbf{k})}|| \leq \texttt{tol}||\mathbf{b}||;$$

the relative error is then

$$\frac{\left\|\mathbf{x}-\mathbf{x}^{(\mathbf{k})}\right\|}{\|\mathbf{x}\|} \leq \kappa\left(\mathbf{A}\right) \frac{\left\|\mathbf{r}^{(\mathbf{k})}\right\|}{\|\mathbf{b}\|} \leq \kappa\left(\mathbf{A}\right) \texttt{tol};$$

the larger $\kappa\left(\mathbf{A}\right)$, the smaller `tol`.

Termination Criterion: Criterion 2

We can also compare the difference between consecutive iterations: $||\mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}||$.

$$
\begin{aligned}
||\mathbf{x} - \mathbf{x}^{(\mathbf{k})}|| &\leq ||\mathbf{T}||||\mathbf{x} - \mathbf{x}^{(\mathbf{k-1})}|| = ||\mathbf{T}||||\mathbf{x} - \mathbf{x}^{(\mathbf{k})} + \mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}|| \\
&\leq ||\mathbf{T}|| \left(||\mathbf{x} - \mathbf{x}^{(\mathbf{k})}|| + ||\mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}|| \right)
\end{aligned}
$$

$$
\Rightarrow ||\mathbf{e}^{(\mathbf{k})}|| = ||\mathbf{x} - \mathbf{x}^{(\mathbf{k})}|| \leq \frac{||\mathbf{T}||}{1 - ||\mathbf{T}||} ||\mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}||;
$$

thus, to decide on $||\mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{k-1})}||$, we need to consider $||\mathbf{T}||/(1 - ||\mathbf{T}||)$.

Stationary Methods: Limitation

T, or better **M**, is not updated at each iteration, producing slow convergence.

Updating **M** is the goal of the Conjugate Gradient Method (**A** is SPD) and Krylov Subspace Methods.

Krylov Subspace Methods, such as GMRES (for general matrices) and MINRES (for symmetric matrices), can handle other classes of matrices.

The Big Picture

We are looking at iterations of the form

$$\mathbf{x}^{(\mathbf{k}+\mathbf{1})} = \mathbf{x}^{(\mathbf{k})} + \alpha^k \mathbf{p}^{(\mathbf{k})}$$

$\mathbf{p}^{(\mathbf{k})} \leftarrow$ search direction
$\alpha^k \leftarrow$ step size

- stationary methods: $\alpha^k = 1$ and $\mathbf{p}^{(\mathbf{k})} = \mathbf{M}^{-\mathbf{1}}\mathbf{r}^{(\mathbf{k})}$;
- gradient descent methods: $\mathbf{p}^{(\mathbf{k})} = \mathbf{r}^{(\mathbf{k})}$; how to choose α^k ?

Steepest Descent Method

Theorem: $\mathbf{x}^*$ produces a minimum of

$$\phi(\mathbf{x}) = \frac{1}{2}\langle \mathbf{x}, \mathbf{A}\mathbf{x}\rangle - \langle \mathbf{x}, \mathbf{b}\rangle$$

if and only if $\mathbf{x}^*$ satisfies the positive definite system $\mathbf{A}\mathbf{x} = \mathbf{b}$.

Sketch of proof:

- $\nabla\phi = \mathbf{A}\mathbf{x} - \mathbf{b} = -\mathbf{r} = 0$;
- the Hessian of ϕ is $\mathbf{A}$, which is SPD, guaranteeing a unique minimizer;

Steepest Descent Method

how to choose α^k?

1. choose α^k that minimizes

$$\begin{aligned}\phi\left(\mathbf{x}^{(\mathbf{k}+\mathbf{1})}\right) &= \phi\left(\mathbf{x}^{(\mathbf{k})} + \alpha^k \mathbf{r}^{(\mathbf{k})}\right) \\ &= \frac{1}{2}\langle\left(\mathbf{x}^{(\mathbf{k})} + \alpha^{\mathbf{k}}\mathbf{r}^{(\mathbf{k})}\right), \mathbf{A}\left(\mathbf{x}^{(\mathbf{k})} + \alpha^{\mathbf{k}}\mathbf{r}^{(\mathbf{k})}\right)\rangle \\ &\quad - \langle\left(\mathbf{x}^{(\mathbf{k})} + \alpha^{\mathbf{k}}\mathbf{r}^{(\mathbf{k})}\right), \mathbf{b}\rangle;\end{aligned}$$

2. differentiate with respect to α^k and set the result to zero:

$$\alpha^k = \frac{\langle \mathbf{r}^{(\mathbf{k})}, \mathbf{r}^{(\mathbf{k})}\rangle}{\langle \mathbf{r}^{(\mathbf{k})}, \mathbf{A}\mathbf{r}^{(\mathbf{k})}\rangle};$$

Steepest Descent Method

- slow convergence: similar to stationary methods;
- preconditioning can help;
- but there are better methods in the class of the gradient descent.

Conjugate Gradient Method (CG)

- published in 1952;
- $\mathbf{A}$ is SPD;
- this method is not as efficient as GE with pivoting: it also requires n steps for a $n \times n$ system, but each step is more costly in terms of computations;
- when is the CG method preferred over GE and the other iterative methods? as an iterative method for solving large sparse systems of equations in $\sqrt{n}$ iterations.

Conjugate Gradient Method: Entities

- energy norm: let $\mathbf{A}$ be SPD; $\mathbf{A}$ induces the norm

$$||\mathbf{x}||_A = \sqrt{\langle \mathbf{x}, \mathbf{A}\mathbf{x} \rangle};$$

- Krylov subspace: let $\mathbf{A}$ be an $n \times n$ real matrix; the Krylov subspace of $\mathbf{A}$ with respect to $\mathbf{b}$ is

$$\mathcal{K}_k\left(\mathbf{A}, \mathbf{b}\right) = \text{span}\left\{\mathbf{b}, \mathbf{A}\mathbf{b}, \mathbf{A}^2\mathbf{b}, \ldots \mathbf{A}^{\mathbf{k}-1}\mathbf{b}\right\}.$$

Conjugate Gradient Method

$$\mathbf{x}^{(\mathbf{k}+\mathbf{1})} = \mathbf{x}^{(\mathbf{k})} + \alpha^k \mathbf{p}^{(\mathbf{k})}$$

- basic idea: minimize the energy norm of the error at each iteration:

$$\min ||\mathbf{e}^{(\mathbf{k})}||_A = \min ||\mathbf{x}^{(\mathbf{k})} + \alpha^k \mathbf{p}^{(\mathbf{k})} - \mathbf{x}||_A;$$

- at $k = 0$, set $\mathbf{x}^{(\mathbf{0})} = \mathbf{0}$, $\mathbf{p}^{(\mathbf{0})} = \mathbf{b} = \mathbf{r}^{(\mathbf{0})}$;
- $\mathbf{x}^{(\mathbf{1})} = \mathbf{x}^{(\mathbf{0})} + \alpha^0 \mathbf{r}^{(\mathbf{0})}$;
- choose the search directions $\mathbf{p}^{(\mathbf{0})} = \mathbf{r}^{(\mathbf{0})}, \ldots, \mathbf{p}^{(\mathbf{n}-\mathbf{1})}$ of n nonzero $\mathbf{A}$-orthogonal vectors, that is,

$$\langle \mathbf{p}^{(i)}, \mathbf{A}\mathbf{p}^{(j)} \rangle = 0 \qquad \text{for } i \neq j;$$

Conjugate Gradient Method

$$\mathbf{x}^{(\mathbf{k}+\mathbf{1})} = \mathbf{x}^{(\mathbf{k})} + \alpha^k \mathbf{p}^{(\mathbf{k})}$$

with

$$\mathbf{p}^{(\mathbf{k}+\mathbf{1})} = \mathbf{r}^{(\mathbf{k}+\mathbf{1})} + \frac{\langle \mathbf{r}^{(\mathbf{k}+\mathbf{1})}, \mathbf{r}^{(\mathbf{k}+\mathbf{1})} \rangle}{\langle \mathbf{r}^{(\mathbf{k})}, \mathbf{r}^{(\mathbf{k})} \rangle} \mathbf{p}^{(\mathbf{k})}$$

and

$$\alpha_k = \frac{\langle \mathbf{r}^{(\mathbf{k})}, \mathbf{r}^{(\mathbf{k})} \rangle}{\langle \mathbf{p}^{(\mathbf{k})}, \mathbf{A}\mathbf{p}^{(\mathbf{k})} \rangle};$$

then, in exact arithmetic,

$$\mathbf{A}\mathbf{x}^{(\mathbf{n})} = \mathbf{b}, \qquad \mathbf{x}^{(\mathbf{k})} - \mathbf{x}^{(\mathbf{0})} \in \mathcal{K}_{\mathbf{k}}(\mathbf{A}, \mathbf{b})$$

Convergence of CG

$$||\mathbf{x} - \mathbf{x}^{(\mathbf{k})}||_A \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k ||\mathbf{x} - \mathbf{x}^{(\mathbf{0})}||_A$$

1. if the original matrix **A** is ill-conditioned, then the conjugate gradient method suffers greatly from the curse of roundoff errors;
2. the solution is to **preconditioning** the system, that is, replace it with one better-conditioned
3. check homework problem.

Preconditioning

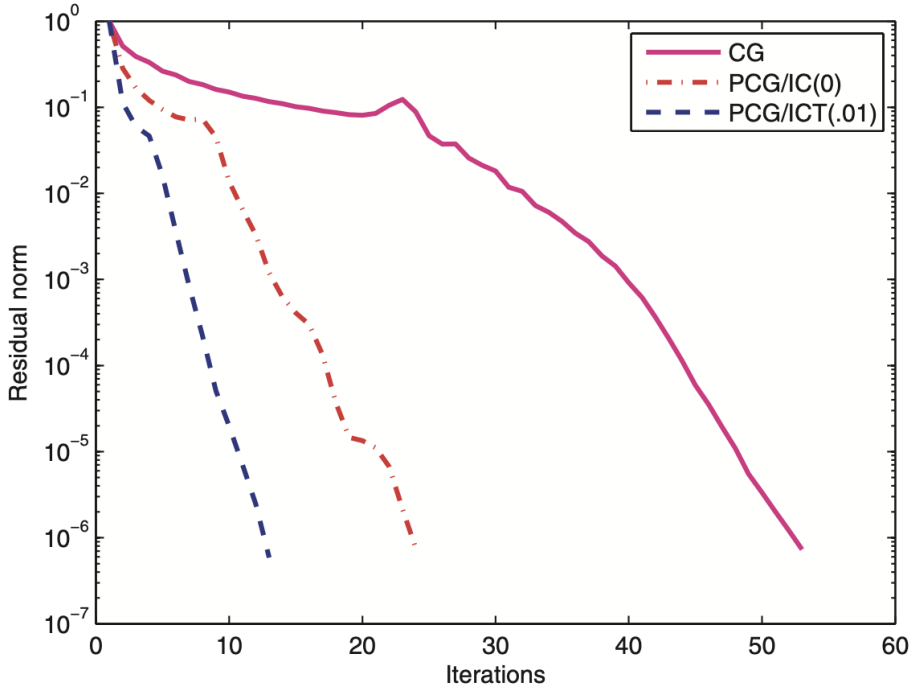
- preconditioning is often used to improve convergence:

$$\mathbf{M}^{-1}\mathbf{A}\mathbf{x} = \mathbf{M}^{-1}\mathbf{b};$$

- $\mathbf{M}$ is the preconditioning matrix, meant to improve the condition number of $\mathbf{M}^{-1}\mathbf{A}$;
- Matlab has a routine, pcg, to solve a system of equations using CG or precondioned CG

Preconditioning

a good preconditioner can produce a substantial improvement in convergence; for example, for a 2d Poisson's equation on a 31×31 grid, with 961 unknowns:



Approximation Theory

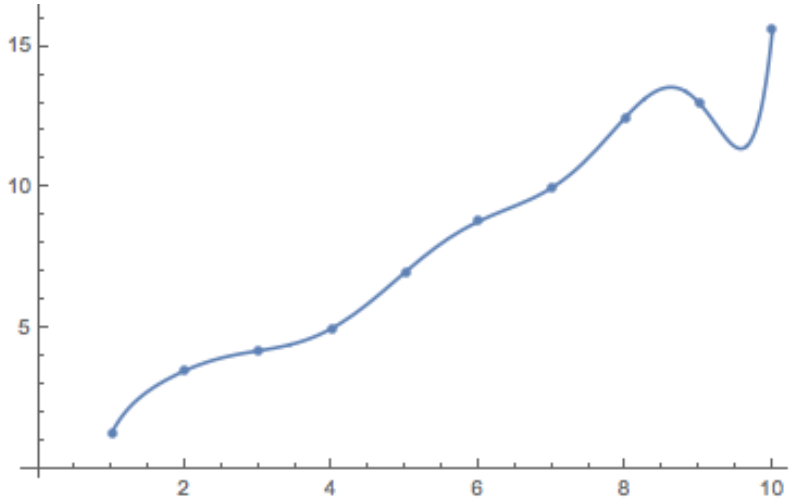
2 classes of problems for approximating a function:

1. the function is known but we want a simpler representation, for example, a polynomial representation;
2. finding the function that best fits a data set.

We will study poly interpolation and splines. In these approaches, the data always agrees with the approximations at the given points. What if the data has errors? It may be unreasonable to ask that the approximating function exactly agrees with the data points.

Example

x_i	1	2	3	4	5	6	7	8	9	10
y_i	1.3	3.5	4.2	5.0	7.0	8.8	10.1	12.5	13.0	15.6



The line in the graphic corresponds to a 9th order poly interp and the dots to the data points. This is poly interpolation and it is not what we are discussing today.

The Best Line

$(x_i, y_i) \rightarrow$ data points

We are looking for the best fitting line through those data points:

$$y = y(x) = cx + d.$$

One good approach to determine a linear approximation (minmax problem, not easily solved):

$$\min_{c,d} \max_{1 \leq i \leq n} |y_i - (cx_i + d)|.$$

Another approach (minimize the absolute deviation):

$$\min_{c,d} \sum_{i=1}^{n} |y_i - (cx_i + d)|.$$

Linear Least Squares

A very convenient approach is

$$\min_{c,d} \sum_{i=1}^{n} \left[y_i - (cx_i + d)\right]^2 ,$$

because:

- it does not put much weight on a piece of badly data, as the minmax approach does;
- it gives sufficient weight to a data point far from the line, which does not happen when the error is written in terms of absolute deviation.

Linear Least Squares

$$\min_{c,d} \phi(c,d) = \min_{c,d} \sum_{i=1}^{n} [y_i - (cx_i + d)]^2$$

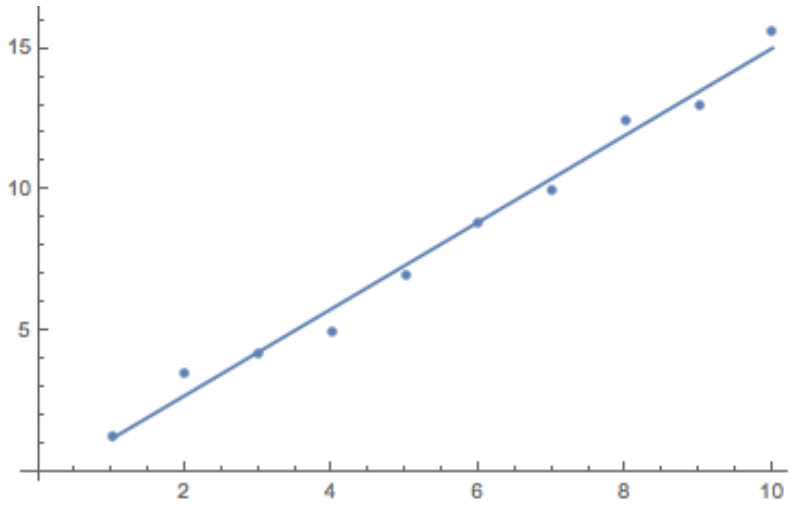
Necessary conditions for the minimum of ϕ with respect to c and d is

$$\frac{\partial \phi}{\partial c} = 0 \qquad \text{and} \qquad \frac{\partial \phi}{\partial d} = 0,$$

which simplify to the **normal equations**:

$$dn + c\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i \qquad \text{and} \qquad d\sum_{i=1}^{n} x_i + c\sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i$$

Back to Example



The line is $y = 1.538x - 0.360$, the linear least squares approximation to the data set.

Polynomial Least Squares

We can construct an approximation for the set of n data points using an algebraic poly

$$P_m(x) = a_m x^m + a_{m-1} x^{m-1} + \ldots + a_1 x + a_0,$$

such that $m < n - 1$, using the same approach:

$$\min_{a_m,\ldots,a_0} \phi(a_m, a_{m-1}, \ldots, a_1, a_0) = \min_{a_m,\ldots,a_0} \sum_{i=1}^{n} [y_i - P_m(x_i)]^2$$

for each $j = 0, \ldots, m$:

$$\frac{\partial \phi}{\partial a_j} = 0$$

so that we have $m + 1$ normal equations, which form a system of $m + 1$ linear equations in a_j.

Linear Algebra Approach

x_i	1	2	3	4	5	6	7	8	9	10
y_i	1.3	3.5	4.2	5.0	7.0	8.8	10.1	12.5	13.0	15.6

$$y_i = cx_i + d$$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \\ 6 & 1 \\ 7 & 1 \\ 8 & 1 \\ 9 & 1 \\ 10 & 1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} c \\ d \end{bmatrix}}_{x} = \underbrace{\begin{bmatrix} 1.3 \\ 3.5 \\ 4.2 \\ 5.0 \\ 7.0 \\ 8.8 \\ 10.1 \\ 12.5 \\ 13.0 \\ 15.6 \end{bmatrix}}_{b}$$

Linear Algebra Approach

- $A_{m\times n}x = b, \quad m > n, \quad A$ has full column rank;
- in general, $Ax = b$ has no solution;
- so, we seek
$$\min_x ||r||_2^2 = \min_x ||b - Ax||_2^2;$$
- the minimizer $\tilde{x}$ exists and is unique since
$$\min_x ||b - Ax||_2^2 = \min_x \left(x^T A^T Ax - 2b^T Ax + b^T b\right)$$
and $A^T A$ is positive definite because A has full column rank;
- $\tilde{x}$ is the least squares approximate solution of $Ax = b$;

Linear Algebra Approach

- $\tilde{x}$ satisfies the normal equations

$$A^T A\tilde{x} = A^T b;$$

- to determine $\tilde{x}$, solve the $n \times n$ system of the normal equations;
- in general, $\tilde{x}$ does not satisfy $Ax = b$:

$$A\tilde{x} \neq b;$$

- solving $Ax = b$ makes $r = 0$;
- least squares attempts to make all the components of r small.

Geometric Interpretation

- let $c_1, c_2, \ldots c_n$ be the columns of A;
- then,

$$||Ax - b||^2 = ||(c_1x_1 + c_2x_2 + \ldots + c_nx_n) - b||^2;$$

- so, in minimizing $||Ax - b||^2$, least squares finds the closest linear combination of the columns of A to the vector b, that is, $A\tilde{x}$ is the closest vector to b in the column space of A;
- rewrite the normal equations as

$$A^T(b - A\tilde{x}) = A^T r = 0;$$

then, the residual of the least squares solution is orthogonal to the column space of A;

Pseudo-Inverse of A

Rewrite the normal equations

$$A^T A\tilde{x} = A^T b$$

as

$$\tilde{x} = \underbrace{\left(A^T A\right)^{-1} A^T}_{\text{pseudo-inverse of} A: A^+} \quad b;$$

$$A^+ = \left(A^T A\right)^{-1} A^T$$

Least Squares Algorithm: Normal Equations

1. build $B = A^T A, \quad y = A^T b$;
2. compute Cholesky factor G of $B = GG^T$;
3. solve $Gz = y$ for z;
4. solve $G^T x = z$ for x.

corresponding to

1. mn^2 flops;
2. $n^3/3 + \mathcal{O}(n^2)$ flops;
3. $\mathcal{O}(n^2)$ flops;
4. $\mathcal{O}(n^2)$ flops.